

Hints and Solutions

MathonGo

Q1 (4)

For a, b, c

$$\text{mean} = \frac{a+b+c}{3} (= \bar{x})$$

$$b = a + c$$

$$\Rightarrow \bar{x} = \frac{2b}{3}$$

$$S.D. (a+2, b+2, c+2) = S.D. (a, b, c) = d$$

$$\Rightarrow d^2 = \frac{a^2+b^2+c^2}{3} - (\bar{x})^2$$

$$\Rightarrow d^2 = \frac{a^2+b^2+c^2}{3} - \frac{4b^2}{9}$$

$$\Rightarrow 9d^2 = 3(a^2 + b^2 + c^2) - 4b^2$$

$$\Rightarrow b^2 = 3(a^2 + c^2) - 9d^2$$

Q2 (5)

$$\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1+n_2} + \frac{n_1n_2}{(n_1+n_2)} (\bar{x}_1 - \bar{x}_2)^2$$

$$n_1 = 10, n_2 = n, \sigma_1^2 = 2, \sigma_2^2 = 1$$

$$\bar{x}_1 = 2, \bar{x}_2 = 3, \sigma^2 = \frac{17}{9}$$

$$\frac{17}{9} = \frac{10 \times 2 + n}{n+10} + \frac{10n}{(n+10)^2} (3-2)^2$$

$$\Rightarrow \frac{17}{9} = \frac{(n+20)(n+10)+10n}{(n+10)^2}$$

$$\Rightarrow 17n^2 + 1700 + 340n = 90n + 9(n^2 + 30n + 200)$$

$$\Rightarrow 8n^2 - 20n - 100 = 0$$

$$2n^2 - 5n - 25 = 0$$

$$\Rightarrow (2n+5)(n-5) = 0 \Rightarrow n = \frac{-5}{2}, 5$$

(Rejected)

Hence n = 5

Q3 (68)

Let number be $a_1, a_2, a_3, \dots, a_{2n}, b_1, b_2, b_3 \dots b_n$

$$\sigma^2 = \frac{\sum a^2 + \sum b^2}{3n} - (5)^2$$

$$\Rightarrow \sum a^2 + \sum b^2 = 87n$$

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Now, distribution becomes $a_1 + 1, a_2 + 1, a_3 + 1, \dots, a_{2n} + 1, b_1 - 1$

$b_2 - 1, \dots, b_n - 1$

Variance

$$\begin{aligned}
 &= \frac{\sum (a+1)^2 + \sum (b-1)^2}{3n} - \left(\frac{12n+2n+3n-n}{3n} \right)^2 \\
 &= \frac{(\sum a^2 + 2n + 2 \sum a) + (\sum b^2 + n - 2 \sum b)}{3n} \\
 &= \frac{(\sum a^2 + 2n + 2 \sum a) + (\sum b^2 + n - 2 \sum b)}{3n} - \left(\frac{16}{3} \right)^2 \\
 &= \frac{87n+3n+2(12n)-2(3n)}{3n} - \left(\frac{16}{3} \right)^2 \\
 &\Rightarrow k = \frac{108}{3} - \left(\frac{16}{3} \right)^2
 \end{aligned}$$

$$\Rightarrow 9k = 3(108) - (16)^2 = 324 - 256 = 68$$

Q4 (35)

$$\frac{\sum x_i}{25} = 40 \& \frac{\sum x_i - 60 + N}{25} = 39$$

Let age of newly appointed teacher is N

$$\Rightarrow 1000 - 60 + N = 975$$

$$\Rightarrow N = 35 \text{ years}$$

Q5 (1)

Let observations are denoted by x_i for $1 \leq i <$

$2n$

$$\bar{x} = \frac{\sum x_i}{2n} = \frac{(a+a+\dots+a)-(a+a+\dots+a)}{2n}$$

$$\Rightarrow \bar{x} = 0$$

$$\text{and } \sigma_x^2 = \frac{\sum x_i^2}{2n} - (\bar{x})^2 = \frac{a^2+a^2+\dots+a^2}{2n} - 0 = a^2$$

$$\Rightarrow \sigma_x = a$$

Now, adding a constant b then $\bar{y} = \bar{x} + b = 5$

$$\Rightarrow b = 5$$

$$\text{and } \sigma_y = \sigma_x \text{ (No change in S.D.)} \Rightarrow a = 20 \Rightarrow a^2 + b^2 = 425$$

Questions with Answer Keys

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Q1: 16 March (Shift 1) - Single Correct

Consider three observations a , b and c such that $b = a + c$. If the standard deviation of $a + 2b + 2, c + 2$ is d , then which of the following is true?

(1) $b^2 = 3(a^2 + c^2) + 9d^2$

(2) $b^2 = a^2 + c^2 + 3d^2$

(3) $b^2 = 3(a^2 + c^2 + d^2)$

(4) $b^2 = 3(a^2 + c^2) - 9d^2$

Q2: 16 March (Shift 2) - Numerical

Consider the statistics of two sets of observations as follows:

	Size	Mean	Variance
Observation I	10	2	2
Observation II	n	3	1

If the variance of the combined set of these two

observations is $\frac{17}{9}$, then the value of n is equal to _____

Q3: 17 March (Shift 2) - Numerical

Consider a set of $3n$ numbers having variance 4. In this set, the mean of first $2n$ numbers is 6 and the mean of the remaining n numbers is 3. A new set is constructed by adding 1 into each of first $2n$ numbers, and subtracting 1 from each of the remaining n numbers. If the variance of the new set is k , then $9k$ is equal to _____.

Q4: 18 March (Shift 1) - Numerical

The mean age of 25 teachers in a school is 40 years. A teacher retires at the age of 60 years and a new teacher is appointed in his place. If the mean age of the teachers in this school now is 39 years, then the age (in years) of the newly appointed

teacher is _____.

Questions with Answer Keys

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Q5: 18 March (Shift 2) - Single Correct

Let in a series of $2n$ observations, half of them are equal to a and remaining half are equal to $-a$. Also by adding a constant b in each of these observations, the mean and standard deviation of new set become 5 and 20, respectively. Then the value of $a^2 + b^2$ is equal to:

(1) 425

(2) 650

(3) 250

(4) 925

Answer Key**Q1 (4)****Q2 (5)****Q3 (68)****Q4 (35)****Q5 (1)**